Solution

SOUND -03

Class 08 - Science

1. **(a)** Above 20000 vibration/sec

   Explanation: Ultrasound applies to all sound energy with a frequency above human hearing (20,000 hertz or 20 kilohertz). Typical diagnostic sonographic scanners operate in the frequency range of 2 to 18 megahertz, hundreds of times greater than the limit of human hearing.

2. **(d)** vocal cord

   Explanation: vocal cord

3. **(b)** 3960 m

   Explanation: 3960 m

4. **(a)** 0.02 s

   Explanation: We know that

   Frequency $= \frac{1}{\text{Time period}}$

   $\Rightarrow$ Time period $= \frac{1}{\text{Frequency}}$

   $\Rightarrow$ Time period $= \frac{1}{50}$ = 0.02 s

5. **(a)** Vibration

   Explanation: The to and fro motion of an object is known as vibration. Vibration is necessary for production of sound.

6. **(a)** Tympanic membrane

   Explanation: The tympanic membrane is the part of ear which vibrates when outside sound fall on it to create vibration.

7. **(b)** Nature of medium

   Explanation: Speed of sound depends upon nature of medium, temperature, humidity, and source of sound but nature of medium influence the speed of sound most. Sound travels faster in solid medium than air and water.

8. **(c)** Electrical Energy

   Explanation: The telephone system converts acoustic energy (sound) into electric energy. The electric energy travels along the phone wires and is converted back into acoustic energy at the receiving end.

9. **(d)** Time period

   Explanation: The time taken by an oscillating pendulum to make one complete oscillation is called its time period. Time taken to complete on vibration is called as time-period of vibrating body.

10. **(a)** Noise

    Explanation: Unpleasant sound is called noise. In classroom, all the students speak together the sound produced will be called noise.

11. **(c)** Auditory nerve

    Explanation: Auditory nerve transmits sound impulses to brain from ear. Optic nerve, cardiac nerves are present in eyes and heart respectively.

12. **(b)** of higher pitch when train arrives

    Explanation: of higher pitch when train arrives

13. **(a)** ultrasonic

    Explanation: ultrasonic

14.

(d) Dogs

Explanation: The police use high frequency whistles which can be heard by dogs but not by human beings.

15.

(c) 60 vibrations per minute

Explanation: 1 = Hz = 1 vibration per second = 60 vibration per minute.

16.

(b) 15 mm

Explanation: 15 mm

17. **(a)** Noise

Explanation: Noise is the sound which is unpleasant to ear. The pleasant sound produced by musical instruments is called music.

18.

(c) Echo

Explanation: The repetition of sound caused by the reflection of sound waves is called Echo. Echo is produced in large halls or in hilly areas due to reflection of sound.

19.

(d) There is no air on the moon

Explanation: Sound requires material medium for its propagation. On the moon, the atmosphere is absent so, sound cannot be heard directly on the moon in absence of air.

20.

(c) ultrasonic waves

Explanation: ultrasonic waves

21.

(c) SONAR

Explanation: SONAR

22.

(b) Solids, liquids and gases

Explanation: Sound require material medium for its propagation. The medium may be solid, liquid or gases. Sound travel faster in solid medium than gaseous and liquid medium.

23.

(c) Difference in amplitude and frequency

Explanation: Different kinds of similar sounds can be differentiated due to difference in amplitude and frequency. Pitch of different persons is also different

24. **(a)** Gas

Explanation: The medium in which sound travels slower is gas.

25. **(a)** Increase in temperature

Explanation: Speed of sound increase with increase in temperature, during propagation of sounds particles of medium travel faster at higher temperature.

26.

(d) Larger amplitude

Explanation: Loudness of sound depends upon amplitude of the sound. Higher the amplitude, louder the sound will be.

27.

(c) SONAR

Explanation: SONAR

28.

(b) Amplify sound

Explanation: Three bones are present in the middle ear named as hammer, anvil, and stirrup. These bones are attached with eardrum one after the other to amplify the sound.

29. **(a)** 0.002 seconds

Explanation: Frequency of sound = 500Hz. Time-period = 1/frequency = 1/500 = 0.002 seconds.

30.

(c) Musical sound

Explanation: The sound which is pleasant to hear is called musical sound while sound which is unpleasant to hear is called noise.

31.

(c) A man

Explanation: The voice of an adult man is of lower pitch in comparison to the voices of a baby boy, a baby girl and a woman. Since frequency of a sound is directly proportional to its pitch, man's voice is of minimum frequency in comparison to a boy, a girl, or awoman's voice.

32.

(b) Square of the amplitude

Explanation: Loudness of sound is proportional to the square of the amplitude. When the amplitude is doubled, loudness increases four times.

33.

(d) Pendulum

Explanation: The motion of object which repeats after equal interval of time is called periodic motion. The motion of Simple pendulum is periodic in nature.

34.

(c) 100 Hz

Explanation: Time-period of a wave is 0.01seconds then

Frequency=1 $\div$ Time period

Frequency= 1 $\div$ 0.01

Frequency= 100 Hz

35.

(b) Male vocal cord is larger

Explanation: Voice of male is heavy compared to woman because male voice has low frequency as compared to female and male vocal cord is larger than female.

36.

(d) Vacuum

Explanation: Sound cannot travel through vacuum as sound require material medium for its propagation. Sound can travel through air, water, and solids.

37.

(d) 0.1s

Explanation: Number of oscillation = 40, time taken = 4 seconds.

Frequency = $40 \div 4$ = 10Hz (number of oscillations made per second)

Time period=1÷Frequency

Time period=1 $\div$ 10 = 0.1s

38.

(b) Temporary or permanent impairment of hearing

Explanation: When a person is exposed to a loud sound continuously he may get temporary impairment of hearing along with other health hazards.

39.

(b) 20 Hz

Explanation: The object oscillating 20 times in one second will have frequency of 20 Hz as number of oscillation or vibration per second is called frequency.

40.

(b) 20 to 20,000 Hz

Explanation: 20 to 20,000 Hz

41.

(d) 40 KHZ

Explanation: 40 KHZ

42.

(c) they can absorb noise.

Explanation: they can absorb noise.

43.

(c) ultrasonic sounds

Explanation: ultrasonic sounds

44.

(b) Wavelength

Explanation: Wavelength

45.

(c) Increase 4 times

Explanation: The loudness of sound depends upon amplitude of sound. When amplitude is doubled than loudness increase by four times.

46.

(b) From birth

Explanation: Total hearing impairment is usually from birth. Hearing impairment may also occurs due to accidents or some other reasons.

47.

(d) above 20,000 Hz

Explanation: Ultrasound has frequency of vibration above above 20000 Hz.

48.

(b) decrease in same proportion

Explanation: If we increase the vibration in an object, the time period will decrease in same proportion. Since, a time period is the time taken for one complete cycle of vibration, i.e.

Time period = $\frac{1}{Frequency}$

49. **(a)** 3300 m

Explanation: Given, speed of sound = 330 m/s and time, t =10 s

We know that, speed = $\frac{distance}{time}$

$\therefore$ distance = speed $\times$ time

So, distance travelled by sound in 10s.

= 330 $\times$ 10 = 3300 m

50.

(c) Loud

Explanation: The loudness of sound depends upon amplitude of sound. So, when the amplitude of sound is large, the sound produced is large.

51.

(c) Frequency is less than 10Hz

Explanation: Human ear can hear sounds of frequency 20Hz to 20,000Hz. So, we are not able to hear the sound of pendulum of frequency less than 10Hz.

52. **(a)** amplitude of vibration

Explanation: Loudness of sound is determined by the amplitude of its vibrations.

53.

(d) Sitar

Explanation: Noise pollution is caused by vehicles, machines, and loudspeakers. Sitar is a musical instrument that does not produce noise.

54.

(b) Damage eardrum

Explanation: We must never put sharp, pointed things into our ear because it can damage eardrum. Eardrum is thin membrane at the junction of outer ear and middle ear.

55.

(b) decibel

Explanation: decibel

56.

(b) Hearing impairment

Explanation: Noise pollution may cause hearing impairment. Unpleasant sound having high altitude may harm the eardrum that leads to hearing problem.

57. **(a)** Diabetes

Explanation: Noise pollution may cause a number of health problem like hypertension, anxiety, and lack of sleep. Diabetes is a hormonal problem not caused due to noise pollution.

58.

(d) Sound waves having frequency below 20 Hz are known as ultrasonic wave

Explanation: Sound waves having frequency below 20 Hz are known as ultrasonic wave

59. **(a)** its amplitude.

Explanation: The maximum displacement from mean position is called its amplitude. The loudness of sound depends upon its amplitude.

60. **(a)** Vibration

Explanation: Sound is produced by the vibration of an object. Vibration creates a disturbance in a medium which reaches to observer's ear through the air, water, or solid medium.

61.

(c) vibration

Explanation: The 'To and fro' or 'back and forth' motion of an object about its mean position, which produced sound is called vibration.

62.

(d) Femur

Explanation: Femur bone is not present in ear. Anvil, Hammer, and stirrups are present in the ears.

63.

(c) Reduce noise pollution

Explanation: Plantation on the road side can reduce noise pollution as plants are good absorber of sound and also reduce the carbon dioxide gas released by vehicles.

64.

(c) generated in a pipe filled with air by moving the piston attached to the pipe up and down

Explanation: generated in a pipe filled with air by moving the piston attached to the pipe up and down

65. **(a)** Steel

Explanation: The speed of sound varies from substance to substance: typically, sound travels most slowly in gases, faster in liquids, and fastest in solids. For example, while sound travels at 343 m/s in air, it travels at 1,481 m/s in water (almost 4.3 times faster) and at 5,120 m/s in iron (almost 15 times faster).

66.

(d) increase its amplitude

Explanation: To increase loudness of sound, increase its amplitude of vibration of sound because loudness of sound is depend on amplitude of vibration of sound.

67.

(d) 2s

Explanation: 2s

68. **(a)** increasing amplitude

Explanation: increasing amplitude

69.

(c) Loudness

Explanation: The intensity or loudness of a sound depends upon the extent to which the sounding body vibrate i.e., the amplitude of vibration. A sound is loudness as the amplitude of vibration is greates. Loudness is measured in units called decibels.

70.

(b) Frequency of 10 Hz

Explanation: Human ear can hear sounds of frequency between 20Hz to 20,000Hz. So, sound of frequency 10Hz cannot be heard by human beings.

71.

(d) Speed = frequency $\times$ wavelength

Explanation: Speed of sound is equal to product of frequency and wavelength. Speed = frequency × wavelength.

72.

(b) Frequency

Explanation: The number of vibration per second is called frequency. The pitch of sound depends upon frequency of sound.

73.

(c) Amplitude is very small

Explanation: The vibrations of object are not visible because in most of the cases amplitude of waves are very small. Amplitude is the maximum displacement from mean position.

74.

(c) Installing industrial activity away from residential area

Explanation: Industrial noise can be reduced by installing industrial activity away from residential area and using safety for workers.

75.

(d) Particles of solid are very close

Explanation: Sound travels faster in the solid medium as compared to liquid and gases because particles of solid are very close to each other.